Open vs Closed AI Models

AI models can be made available in different ways. Some models provide access to their trained weights, while others are available mainly through a provider's application or API.

These differences affect how developers' access, customize, deploy, and maintain AI models. They can also influence infrastructure requirements, performance, and cost.

❖ Open-Weight AI Models

- ### Understanding Open-Weight and Open-Source AI Models

1. Open-Weight AI Models

An open-weight AI model is an AI model whose trained weights are available for people to access and use, depending on its license.

What are weights?
Weights are numbers that an AI model learns during training. These numbers help the model recognize patterns and give answers.

How do weights work?
For example, suppose we give an AI model a picture of a dog. The model uses its existing weights to understand the picture and predict that it contains a dog.
We do not need to give the weights separately every time we provide an image. The model already has its trained weights.

Can we improve the model?
Yes. If the model makes mistakes while identifying dogs, we can train it further using correctly labeled dog images. This process is called *fine-tuning*. It updates the model's weights and may help it make better predictions.
Example:
Some Qwen models allow developers to download their trained weights and run them on their own computers or servers, provided they meet the model's requirements and license terms.

Key Takeaway: Open-weight models allow developers to access their trained weights. They can use these models and, where permitted, fine-tune them for specific tasks.

2. Open-Source AI Models

An open-source AI model is a model that allows people to use, study, modify, and share its available components according to its license.
How does it work?
Developers can access the model's available code and other components to understand how it works. Depending on the model and its license, they may also modify the code or share their changes with others.
Example:
The **BLOOM language model**, developed through the BigScience research collaboration, is an example of an openly released language model. Its publicly available resources allow researchers and developers to explore and use the model under its applicable terms.

Key Takeaway: Open-source AI models give developers the freedom to study, use, and modify the available components under the applicable license. However, not every model provides all its training data or other components.

❖ Closed/Proprietary AI Models

A closed or proprietary AI model is an AI model whose trained weights are not publicly available for people to download and use freely. These models are generally accessed through a company's website or API.

The company that provides the model manages its underlying technology and controls how it can be accessed and used.

How do closed models work?
Users send their questions or instructions to the model through a website or API. The model processes the input and returns a response. Users do not need to download or manage the model's underlying weights.

Real-World Example:
ChatGPT is an example of a service that provides access to OpenAI's proprietary GPT models. Users can ask questions and receive answers through the ChatGPT interface without downloading the model's weights.
Similarly, Anthropic provides access to its proprietary Claude models through its services and APIs.

Key Takeaway: Closed/proprietary models allow users to access AI capabilities through provider-managed services, but their underlying trained weights are generally not publicly available.

❖ Model Access

Model access refers to how users connect to and use an AI model.

Open-Weight Models: These models can often be downloaded and run on a local computer or server, depending on their license and hardware requirements. They can also be accessed through APIs or third-party platforms.

Closed/Proprietary Models: These models are generally accessed through the provider's website or API. The provider manages the underlying infrastructure when offering a hosted service.

Example: A developer can run a suitable open-weight model locally or connect an application to OpenAI's API to generate responses.

❖ Customization and Fine-Tuning

Customization means adapting an AI model to meet specific requirements.

Prompting: Providing instructions to guide the model's responses without changing its trained weights.

Fine-Tuning: Training an existing model further using specific examples to improve its performance on a particular task. This process updates some or all of its weights.

Open-weight models may support fine-tuning using suitable tools and hardware, subject to their licenses. Closed models may also support fine-tuning if the provider offers that capability.

Example: A company can fine-tune a suitable AI model using customer-support examples to help it answer common customer questions.

❖ API-Based Deployment vs Self-Hosting

Deployment means making an AI model available for people or applications to use. Hosting means where the AI model runs.

1. Open-Weight Models

Open-weight models can often be downloaded and run on your own computer or server, depending on the model's license and your computer's capabilities. They can also be used through APIs.

Example: A developer downloads a suitable Qwen model and runs it on their own computer.

2. Closed/Proprietary Models

Closed models are generally used through the company's website or API. The company manages the servers that run the model when providing a hosted service. Users usually cannot download the model's weights.

Example: A developer uses OpenAI's API to add AI responses to an application without running the model on their own server.